AgriLens

Dust or Drought?

Satellite evidence for deciding which field to inspect first.

Precision Agriculture & Crop Intelligence

Real Sentinel-2 histories and NASA EMIT spectra
Nile Delta, Egypt

The agricultural decision

A change in greenness has several possible causes

Water stress, harvest, dust or atmospheric contamination can produce similar-looking changes in satellite measurements.

The user needs evidence before acting

An agronomist or irrigation manager reviews the flagged field, checks the imagery and asks for field observations before changing management.

Our PoC prioritises inspection. Field evidence determines the cause.

Scope: README.md and 06_Documentation/DATASET_CARD.md

The evidence we actually have

One approximately 3 × 3 km Egyptian study area, March-September 2025

615

field-date rows

20

audited field units

33

retained field dates

Why Egypt?

The committed area has a usable time series and real local EMIT spectra, enabling a reproducible regional demonstration.

What this does not establish

A Jordan pilot still needs local data, field references and evaluation. We do not claim that Jordan lacks hyperspectral coverage.

Inputs: validation/data/agrilens_real_field_date_QA_gated_615.csv; 03_EMIT_hyperspectral/; source provenance

The current modeling workflow

Forecast-residual method v1.2, separate from the dashboard median-deviation screen

01 Historical features

Lagged NDVI and change, observation gaps, season, NDMI and modeled environmental context.

02 Forecast and compare

Fit on training fields, select on validation fields, compare with a persistence baseline.

03 Quality review

Apply a training-only residual threshold, then hold suspected scene effects separately.

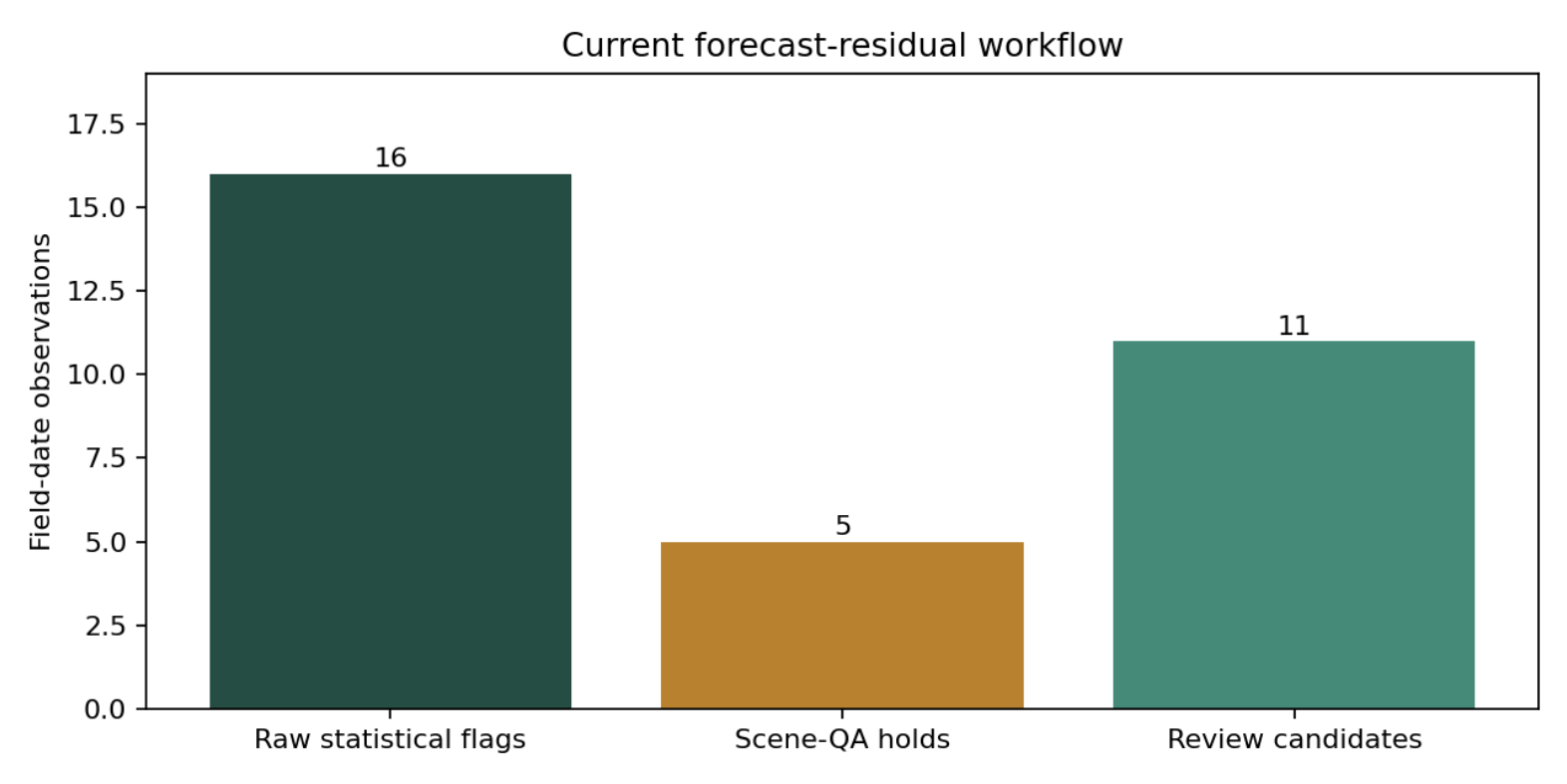
04 Field inspection

Export a review queue and geographic points for human investigation.

Implementation: scripts/ndvi_anomaly_detection.py; reproduced by AgriLens_PoC.ipynb

The reproduced screening result

All geographic splits combined. Counts refer to field-date observations.



16 raw flags

5 receive a quality hold on 17 May 2025. 11 remain as inspection candidates.

No verified damage labels.
No causal diagnosis.

New execution: output/notebook_run/model_run_manifest.json. Earlier 15 / 5 / 10 counts belong to an older feature setup.

The working dashboard

Actual screenshot of the bundled interactive demonstration



Dashboard: docs/index.html. Its 23 observations include 14 positive and 9 negative deviations.

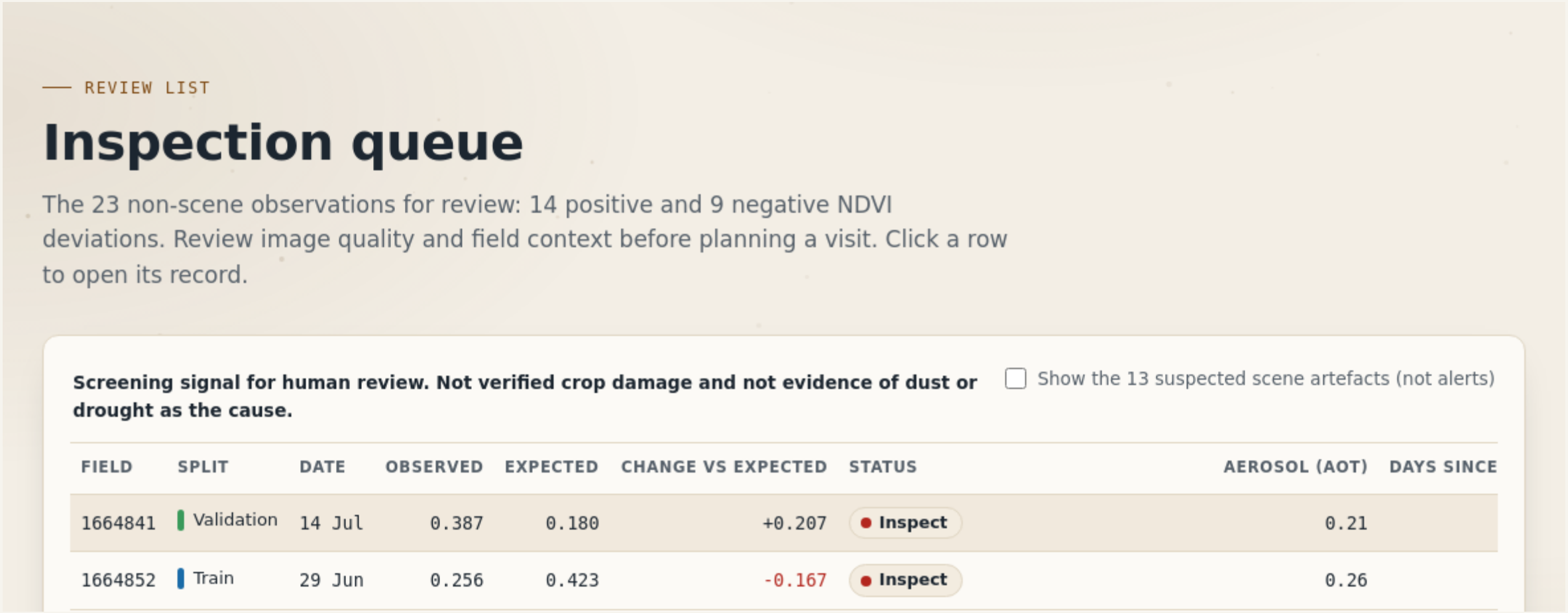
The dashboard shows the separate validation workflow:

36 raw deviations
13 scene holds
23 review observations

Open docs/index.html
Works offline.

The review list supports field follow-up

Actual dashboard screenshot. This is the median-deviation validation workflow.



— REVIEW LIST

Inspection queue

The 23 non-scene observations for review: 14 positive and 9 negative NDVI deviations. Review image quality and field context before planning a visit. Click a row to open its record.

Screening signal for human review. Not verified crop damage and not evidence of dust or drought as the cause.

☐ Show the 13 suspected scene artefacts (not alerts)

FIELD	SPLIT	DATE	OBSERVED	EXPECTED	CHANGE VS EXPECTED	STATUS	AEROSOL (AOT)	DAYS SINCE
1664841	Validation	14 Jul	0.387	0.180	+0.207	Inspect	0.21	
1664852	Train	29 Jun	0.256	0.423	-0.167	Inspect	0.26	

Each row is an observation to review. The list does not prove crop damage or prescribe irrigation.

Validation against a simple baseline

Current model: 9 training / 6 validation / 5 held-out test fields, 152 test pairs

Held-out method	MAE	RMSE
Persistence	0.029575	0.043602
Ridge (alpha 1.0)	0.031036	0.040376

What the comparison shows

Persistence has lower MAE. Ridge has lower RMSE. We selected Ridge on validation fields, not on the held-out test.

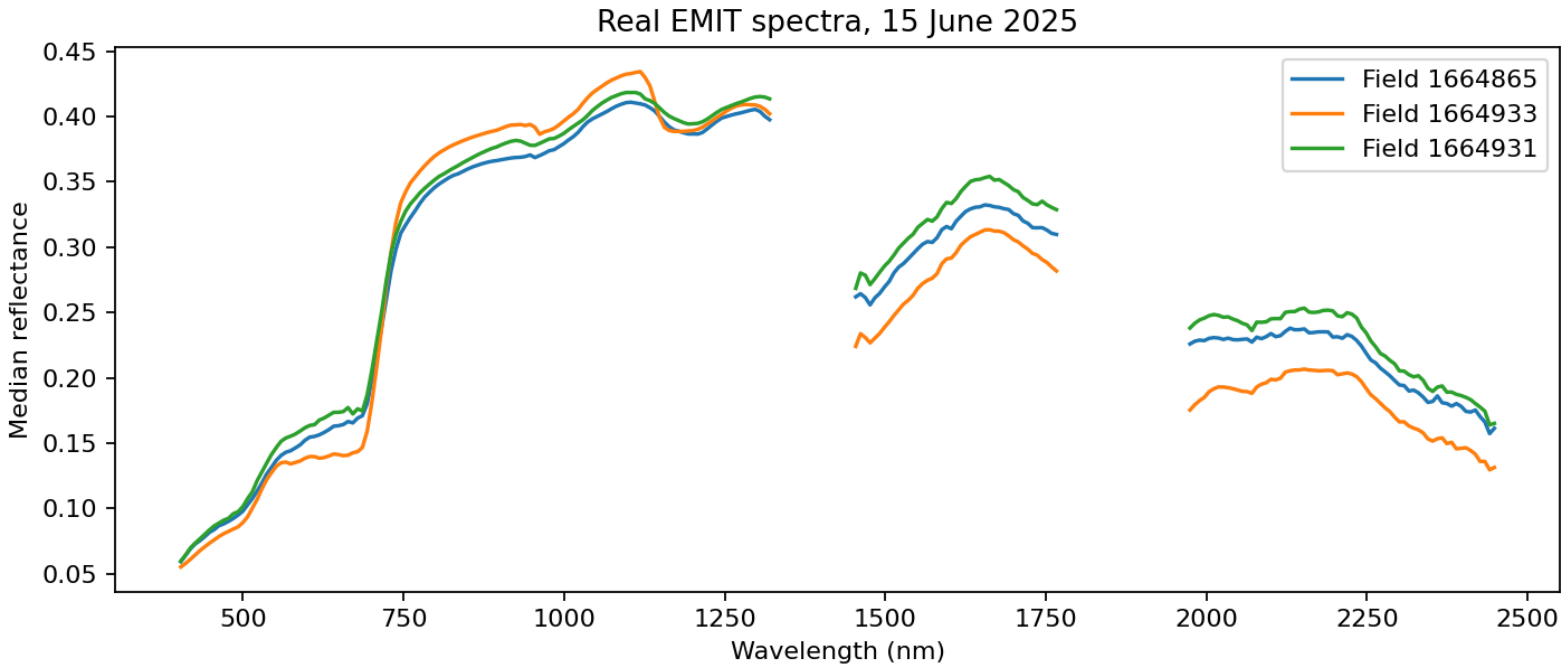
What the checks prove

13 regression tests pass. These verify software behavior, including split and threshold controls. They do not establish agronomic accuracy.

Results: output/notebook_run/forecast_metrics.csv. The archived 37-pair benchmark is a different experiment.

Real hyperspectral observations

NASA EMIT, 15 June 2025. Paired Sentinel-2 observation: 16 June 2025.



14 usable fields
232 complete wavelengths
3,248 field-band rows

Six fields lack usable spectra. Coverage differs across the geographic splits.

Exploratory spectra. EMIT is not a feature input to the temporal forecast.

Source: NASA EMIT L2A RFL, doi.org/10.5067/EMIT/EMITL2ARFL.001. No Satellite 813 observations are claimed.

Limits and the next evidence needed

Current limits

One small area and one season. Modeled environmental context. Predicted field boundaries. Uneven spectral coverage. No independently verified damage labels.

Next validation

Collect dated agronomist observations, crop and irrigation records, and field moisture measurements. Test a separate season and region. Compare with persistence again.

Water savings, yield improvements and dust/drought classification accuracy remain unmeasured.

Evidence boundaries: 06_Documentation/DATASET_CARD.md and README.md

Pilot, impact and incubation roadmap

Proposed next steps and business hypothesis

Month 1: partner and references

Recruit an agricultural adviser or cooperative. Agree on field labels, review steps and farmer consent for pilot records.

Months 2-3: evaluate in use

Measure review time, useful-review rate against independent labels and performance on new fields and dates.

Service hypothesis

A monitoring subscription for agricultural advisers or cooperatives, starting with inspection prioritisation.

Incubation request

Access to farm partners, agronomic mentors and repeated hyperspectral observations. Test willingness to pay before setting prices.

Proposals only. No signed pilot partner, commercial revenue or measured resource saving is claimed.

Team and source credits

Maya - data acquisition and QA
Rand - NDVI modeling
Rama - validation and geographic splits
Nour - dashboard and figures
Omar - documentation and presentation

Copernicus Sentinel-2 (2025)
NASA EMIT / LP DAAC
Fields of the World / Source Cooperative
Open-Meteo / CAMS context
ESA WorldCover (2021)

Reproducible entry point: AgriLens_PoC.ipynb

Full product links, licences and modifications:
DATA_SOURCES_AND_ATTRIBUTION.md

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